Unit 10: Nonhomogeneous Equations and Forced Response

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Condensed Cheat Sheet: Particular Solutions and Forced Response

Unit 10 at a Glance: Core Sub-Topics

A nonhomogeneous linear equation reads $L[y] = g(x)$, and its general solution always splits as $y = y_h + y_p$ — the complementary solution of $L[y] = 0$ plus *any* particular solution of the forcing. This unit is the toolkit for building y_p and for reading how a system answers an input.

- **Undetermined coefficients** — guess a trial of the same shape as the forcing, then match coefficients.
- **Variation of parameters** — promote the constants c_1, c_2 to functions for *any* continuous forcing.
- **The annihilator method** — an operator that kills g derives the trial form instead of memorizing it.
- **Transient and steady-state** — the homogeneous part decays, the particular part persists.
- **Unit step and impulse response** — $u(t)$, $\delta(t)$, and the system's answer to a unit kick from rest.
- **Duhamel's principle** — convolve the impulse response with the input to answer *every* forcing at once.

Part I — Condensed Cheat Sheet

10.1 The Method of Undetermined Coefficients

For a constant-coefficient $L[y] = g(x)$, the general solution is $y = y_h + y_p$: the complementary solution y_h solves $L[y] = 0$ and carries the arbitrary constants, while y_p matches the forcing. When g is an exponential, polynomial, sine/cosine, or a product of these, guess a trial of the *same shape* and solve for its coefficients.

Complementary Plus Particular

$$y = y_h + y_p, \qquad L[y_h] = 0, \qquad L[y_p] = g(x).$$

Only y_h holds the free constants, so it alone can absorb the initial or boundary conditions — which is why y_p is never the whole story.

Trial Solution Table (No Overlap)

Forcing $g(x)$	**Trial** y_p
$e^{\alpha x}$	$Ae^{\alpha x}$
polynomial of degree n	$A_n x^n + \cdots + A_1 x + A_0$
$\cos \beta x$ or $\sin \beta x$	$A \cos \beta x + B \sin \beta x$
$e^{\alpha x} \cos \beta x$	$e^{\alpha x}(A \cos \beta x + B \sin \beta x)$

For a *sum* of forcings, build a particular solution for each piece and add them (superposition for the nonhomogeneous equation).

Common Pitfall: Overlap with the Homogeneous Solution

If any term of the trial already solves $L[y] = 0$, substitution collapses it to zero and the method fails. **Multiply the offending trial by** x (or by a higher power x^k for a root of multiplicity k) until no term overlaps y_h. For $y'' + y = \cos x$, the bare guess $A \cos x + B \sin x$ *is* the homogeneous solution; the corrected resonant trial is $x(A \cos x + B \sin x)$.

10.2 Variation of Parameters

When the forcing is not one of the friendly shapes — $\tan x$, $\sec x$, e^x/x — undetermined coefficients has nothing to guess. Variation of parameters promotes the constants c_1, c_2 of $y_h = c_1 y_1 + c_2 y_2$ to functions $u_1(x), u_2(x)$ and works for *any* continuous g.

Variation of Parameters Formulas

Write the equation in standard form $y'' + py' + qy = g$ (leading coefficient one), let $W = y_1 y_2' - y_2 y_1'$ be the Wronskian, and set $y_p = u_1 y_1 + u_2 y_2$ with

$$u_1' = -\frac{y_2\, g}{W}, \qquad u_2' = \frac{y_1\, g}{W},$$

$$y_p = -\, y_1 \int \frac{y_2\, g}{W}\, dx \; + \; y_2 \int \frac{y_1\, g}{W}\, dx.$$

Common Pitfall: Forgetting to Normalize

The g in these formulas is the right-hand side *after* dividing through so the y'' coefficient is one. Reading g off an unnormalized equation scales the answer incorrectly. Always normalize first, then read g.

10.3 Differential Operators and the Annihilator Method

Writing $D = \frac{d}{dx}$, a constant-coefficient operator factors like a polynomial. An **annihilator** of g is an operator $A(D)$ with $A(D)g = 0$. Applying it to both sides of $L[y] = g$ turns the problem into a larger *homogeneous* equation whose extra solutions are exactly the trial form.

Annihilator Workflow

1. Factor $L(D)$ and write the complementary solution y_h.
2. Pick the annihilator $A(D)$ of the forcing g.
3. Apply $A(D)$: solve the homogeneous equation $A(D)L(D)\,y = 0$.
4. Drop the terms already in y_h; the survivors are the trial y_p.
5. Substitute into $L[y] = g$ and match coefficients.

Common Annihilators

Function	**Annihilator**
$e^{\alpha x}$	$(D-\alpha)$
x^{k-1}	D^k
$\cos\beta x,\ \sin\beta x$	$(D^2+\beta^2)$
$e^{\alpha x}\cos\beta x,\ e^{\alpha x}\sin\beta x$	$(D-\alpha)^2+\beta^2$

A repeated factor $(D-\alpha)^k$ signals the terms $e^{\alpha x}, xe^{\alpha x}, \ldots, x^{k-1}e^{\alpha x}$. Annihilate a sum by *multiplying* the individual annihilators.

10.4 Transient and Steady-State Response

For a *stable* forced system (characteristic roots with negative real parts), the complete solution $y = y_h + y_p$ separates by long-term behavior: the homogeneous part fades while the particular part endures.

Transient Versus Steady-State

$$y(t) = \underbrace{y_h(t)}_{\text{transient } \to\ 0} + \underbrace{y_p(t)}_{\text{steady-state}}.$$

The **transient** carries the initial-condition dependence and decays because its exponentials have negative real parts. The **steady-state** is fixed by the forcing alone, so long after the start the response is essentially independent of the initial data. For sinusoidal forcing the steady state is a sinusoid at the *drive* frequency, rescaled and phase-shifted, often read off the complex gain $1/p(i\omega)$.

10.5 Unit Step and Impulse Response

Two idealized inputs model switching and striking. The unit step $u(t)$ switches a forcing on; the Dirac delta $\delta(t)$ is an instantaneous unit kick. They are linked by calculus, and so are the responses they produce.

Step, Delta, and Their Responses

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}, \qquad \delta(t) = u'(t), \qquad \int_{-\infty}^{\infty} \delta(t)\,dt = 1.$$

Sifting property: $\displaystyle\int_{-\infty}^{\infty} f(t)\,\delta(t-a)\,dt = f(a)$. The **impulse response** $w(t)$ is the output of $L[y] = \delta(t)$ from rest; the **step response** is its integral. Knowing w unlocks every input by convolution.

Common Pitfall: Forgetting the Rest State

The impulse response is defined from **rest** — zero state just before the kick. Starting with leftover energy mixes in a transient that is unrelated to the impulse, so the clean $w(t)$ (here $w(0) = 0$, $w'(0) = 1$ for a second-order system) is lost. Isolate the impulse by zeroing the initial conditions first.

10.6 Duhamel's Principle

Duhamel's principle views a general input as a continuous succession of small impulses. Each kick at time τ produces a scaled, time-shifted copy of the impulse response, and superposition sums them into a convolution.

Duhamel's Convolution

For a linear, time-invariant system starting from rest, the forced response to an input $f(t)$ is

$$y_p(t) = \int_0^t w(t-\tau)\,f(\tau)\,d\tau,$$

where w is the impulse response (the Green's function of the initial-value problem). The limits 0 to t encode **causality** — only inputs already applied can affect the present — and the argument $t - \tau$ is the time elapsed since each kick. **Linearity** and **time invariance** are exactly what make the convolution valid.

Unit 10 Key Takeaway

Every forced response is $y = y_h + y_p$. Build y_p by undetermined coefficients when the forcing is friendly (remember to multiply by x on overlap), by variation of parameters for anything continuous, or read its form from an annihilator. In a stable system the transient y_h decays and the steady-state y_p remains, and once the impulse response w is known, Duhamel's convolution answers every input at once.